

Generation Interconnection System Impact Study Report

For Queue Project Q18-1120

Welco 138 kV
January 23, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1120. The Interconnection Customer proposes a 334.4 MW wind facility interconnecting at the Welco 138 kV substation in the Virginia area. The request was received on December 05, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 280 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$99,408,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mecklenburg Power Sta 345 kV/138 kV Autotransformer	Mecklenburg Power Sta – Person 345 KV	151.2	383	16.3
Cheboygan – Trinidad 138 KV	Mecklenburg Power Sta – Person 345 KV	150.9	232	8.7
Welco – Sinai 138 KV	Person – Mecklenburg Power Station 345 KV	138.8	318	41.2
Halifax – Sinai 138 KV	Vewind – Sinai 138 KV	134.2	219	37.6
Cape Horn – Allied 138 KV	Bear – Clover 345 KV	130.9	223	5.6
Halifax – Venwind 138 KV Line	Venwind – Sinai 138 KV	124.4	231	28.7
Trinidad – Brimley Substation 138 KV	Person – Mecklenburg Power Station 345 KV	123.8	209	7.0
Skylark – Allied 138 KV	Bear – Clver 345 KV	119.4	230	5.6
Venwind – Huben 138 KV	Clover – Bear 345 KV	118.5	339	32.6
Sinai – Venwind 138 KV	Halifax – Sinai 138 KV	113.5	264	37.3
Pine River Alte – Welco Substation 138 KV	Welco – Sinai Substation 138 KV	113.1	282	100.0
Mecklenburg Power Sta – Cheboygan 138 KV	Mecklenburg Power Station – Person 345 KV L	108.6	346	11.9
Skylark – Corson 138 KV	Bear – Clover 345 KV	107.3	250	5.6
Pine River Alte – State School 138 KV	Welco – Sinai 138 KV	106.7	381	56.2
Corson – Crewe 138 KV	Bear – Clover 345 KV	106.4	248	5.6
Trinidad – State School 138 KV line	Mecklenburg Power Station – Person 345 KV	106.3	501	6.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
12Th Ave Substation – Yates Dam Substation 138 KV line	Hutchins – Bauxite 138 KV	101.0	158	5.1
Jetersville – Crewe Substation 138 KV	Bear – Clover 345 KV	100.9	240	5.6
Huben – Cape Horn 138 KV	Bauxite – Hutchins 138 KV	100.2	356	13.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mecklenburg Power Sta 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$9,668,000	19% (\$1,834,000)
Tinidad – Cheboygan 138 KV	Rebuild line with higher-capacity conductor	\$12,647,000	93% (\$11,775,000)
Welco – Sinai 138 KV	Rebuild line with higher-capacity conductor	\$6,330,000	57% (\$3,637,000)
Halifax – Sinai 138 KV	Rebuild line with higher-capacity conductor	\$5,050,000	57% (\$2,874,000)
Cape Horn – Allied 138 KV Line	Rebuild line with higher-capacity conductor	\$18,611,000	91% (\$16,967,000)
Venwind – Halifax 138 KV line	Rebuild line with higher-capacity conductor	\$3,457,000	51% (\$1,762,000)
Trinidad – Brimley Substation 138 KV	Rebuild line with higher-capacity conductor	\$13,539,000	93% (\$12,543,000)
Skylark – Allied 138 KV	Rebuild line with higher-capacity conductor	\$12,991,000	91% (\$11,844,000)
Huben – Venwind 138 KV Line	Rebuild line with higher-capacity conductor	\$2,955,000	50% (\$1,489,000)
Sinai – Venind 138 KV	Rebuild line with higher-capacity conductor	\$3,972,000	57% (\$2,266,000)
Pine River Alte – Welco Substation 138 KV	Rebuild line with higher-capacity conductor	\$11,414,000	100% (\$11,414,000)
Cheboygan Substation – Mecklenburg Power Sta 138 KV	Reconductor line	\$4,956,000	85% (\$4,223,000)
Skylark – Corson 138 KV L	Reconductor line	\$4,510,000	91% (\$4,111,000)
Pine River Alte – State School 138 KV	Reconductor line	\$1,224,000	39% (\$481,000)
Corson – Crewe 138 KV	Reconductor line	\$1,606,000	91% (\$1,465,000)
Trinidad – State School 138 KV line	Reconductor line	\$1,795,000	76% (\$1,357,000)
12Th Ave – Yates Dam 138 KV L	Reconductor line	\$3,297,000	91% (\$2,992,000)
Jetersville – Crewe 138 KV	Reconductor line	\$3,496,000	91% (\$3,187,000)
Huben – Cape Horn 138 KV	Reconductor line	\$3,867,000	82% (\$3,187,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$99,408,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.